

Unit 1.3

Functions (p. 8-21)

Note Title

2/8/2018

Definition: (p. 8)

A function f is a rule that assigns to each element x in a set D exactly one element $f(x)$ in a set E .

Eg: Area of circle:

4 ways to present a function:

- ① verbally (in words)
- ② numerically (table)
- ③ visually (graph)
- ④ algebraically

Finding the domain and range of a function:

Domain:

Range:

Examples. Find the domain and range of

① $f(x) = \frac{1}{x}$

Domain:

Range:

② $f(x) = \sqrt{x-1}$

Domain:

Range:

③ $f(t) = \frac{1}{t^2}$

Domain:

Range:

④ $f(x) = 2^x$

Domain:

Range:

Piecewise defined functions:

•
$$f(x) = \begin{cases} 2 & \text{if } x < -2 \\ -x^2 + 1 & \text{if } -2 \leq x \leq 1 \\ -x & \text{if } x > 1. \end{cases}$$

Domain

Range:



• $f(x) = x + |x+1|$

Find the domain and range of f . Sketch.

Vertical line test: To test if f is a function, using its graph.

Symmetry of functions: Even/odd

Even function: Symmetric about the Y-axis.

$$f(-x) = f(x) \text{ for all } x \in D.$$

Examples:

$$y = x^2$$

$$y = \cos x$$

Odd function: Symmetric about the origin.

$$\Rightarrow f(-x) = -f(x) \text{ for all } x \in D$$

Examples: Consider

① $f(x) = x^3$

② $f(x) = \sin x$

More examples: f odd, even or neither?

(a) $f(x) = 2x - x^2$

(b) $f(x) = x/|x|$

Increasing / Decreasing functions (p. 16)

- f is increasing on I if
- f is decreasing on interval I if